

University of Technology

Department of Computer Science & Engineering

Internal Assessment 1

Subject: Database Management Systems

Date: 2026-05-22

Max Marks: 50

Duration: 2 Hours

Instructions:

1. Answer all questions.
2. Marks for each question are indicated on the right.
3. Draw neat diagrams wherever necessary.

Module 1

- Q1.** Discuss the role and responsibilities of a Database Administrator. **[5 Marks]**
- Q2.** Explain the three-level architecture of DBMS with a suitable diagram. **[10 Marks]**

Module 2

- Q3.** Explain Entity, Attribute, and Relationship in ER modeling with examples. **[5 Marks]**
- Q4.** Define Candidate Key, Primary Key, Foreign Key, and Super Key with examples. **[10 Marks]**
- Q5.** Convert the given ER diagram into relational schema and identify functional dependencies. **[10 Marks]**

Module 4

- Q6.** What are anomalies in unnormalized relations? Explain insertion, deletion, update anomalies. **[5 Marks]**

Module 5

- Q7.** Explain the Two-Phase Locking (2PL) protocol and its variants. **[5 Marks]**

CO-PO Mapping:

Q.No	Question	Marks	CO	PO	Module
1	Discuss the role and responsibilities of a Databas...	5	CO1	PO1,PO3	1
2	Explain the three-level architecture of DBMS with ...	10	CO1	PO1,PO2	1
3	Explain Entity, Attribute, and Relationship in ER ...	5	CO2	PO1	2
4	Define Candidate Key, Primary Key, Foreign Key, an...	10	CO2	PO1,PO2	2
5	Convert the given ER diagram into relational schem...	10	CO2	PO1,PO2,PO3	2
6	What are anomalies in unnormalized relations? Expl...	5	CO4	PO1	4
7	Explain the Two-Phase Locking (2PL) protocol and i...	5	CO5	PO1,PO2	5
Total		50			